Do Energy Labels Move Markets? Multi-Cutoff Evidence from English Property Transactions

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Abstract

Energy Performance Certificate labels create discrete jumps in information at continuous score thresholds, generating natural experiments for valuing energy efficiency in housing markets. Exploiting five EPC band boundaries as a multi-cutoff sharp regression discontinuity design on 85,795 matched English property transactions (January 2015 through December 2024), I estimate the capitalization of energy labels into transaction prices. Crossing the E/F boundary—where Minimum Energy Efficiency Standards prohibit renting below band E—raises prices by 6.5% ($p = 0.001$). A cross-cutoff decomposition suggests that the purely informational effect of crossing a band boundary is small, implying that the E/F premium is predominantly regulatory—a finding reinforced by the owner-occupied placebo. The energy crisis of 2021–2023 amplified label premia: the E/F discontinuity rose to 15.3% during peak gas prices. Owner-occupied properties show near-zero discontinuities at E/F, confirming that regulatory exposure, not unobserved sorting, drives the rental premium.

JEL Codes: Q48, R31, D83, L51

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1. Introduction

A homebuyer in England receives a color-coded letter grade—from A (most efficient) to G (least efficient)—on the Energy Performance Certificate attached to every property listing. Two homes with nearly identical energy consumption can receive different letter grades if their continuous scores fall on opposite sides of a threshold. Does crossing that threshold change what buyers will pay?

The EPC system offers a rare opportunity to disentangle the information and regulation channels in housing markets (Eichholtz et al., 2010; Fuerst et al., 2015; Aydin et al., 2020; Akerlof, 1970; Leland, 1979). It combines a continuous score with discrete labels—and one particular boundary (E/F) carries regulatory consequences through the Minimum Energy Efficiency Standards (MEES), while others are purely informational.

This paper estimates the causal effect of EPC band assignment on property transaction prices in England using a multi-cutoff sharp regression discontinuity design. The running variable is the Standard Assessment Procedure (SAP) energy efficiency score, which ranges from 1 to 100 and determines band assignment at five thresholds: 39 (E/F), 55 (D/E), 69 (C/D), 81 (B/C), and 92 (A/B). I link the universe of EPC assessments to HM Land Registry Price Paid Data, constructing a matched sample of 85,795 transactions from January 2015 through December 2024.

The multi-cutoff design yields three sets of results. First, I estimate the overall price discontinuity at each boundary. The E/F boundary shows the largest and most statistically significant jump: properties scoring just above 39 (band E) sell for 6.5 percent more than those scoring just below (band F), with a robust bias-corrected standard error of 2.2 percentage points ($p = 0.001$). The A/B boundary also shows a substantial premium of 13.6 percent, while the D/E boundary shows a negative discontinuity of -3.5 percent (significant at the 5% level using bias-corrected inference). The C/D and B/C boundaries yield estimates that are smaller and not statistically distinguishable from zero.

Second, I exploit the unique institutional feature of MEES to decompose the E/F effect into informational and regulatory components. Since April 2018, MEES has prohibited landlords from letting properties rated below band E, making the E/F boundary the only one carrying regulatory force. Using the D/E and C/D boundaries as "informational placebo" cutoffs—where crossing a band boundary changes the label but carries no regulatory consequence—I find that the informational effect at the cleanest non-regulatory boundary (C/D) is small (1.5 percentage points, not significant), suggesting that the bulk of the E/F premium is attributable to MEES regulation. The decomposed regulatory component of 7.5 percentage points ($p = 0.009$) slightly exceeds the total E/F effect because the informational benchmark

is marginally negative, suggesting that buyers respond primarily to the regulatory signal rather than the color of the label per se.

Third, I test whether the UK energy crisis of October 2021 through mid-2023 amplified these label effects. Theory predicts that when energy costs are salient, buyers should place greater value on efficiency—and by extension, on the labels that signal it. I find strong evidence of crisis amplification: the E/F discontinuity roughly doubled during the crisis period, rising from near zero in the post-MEES pre-crisis period to 15.3 percent ($p = 0.016$). The decomposition reveals that this amplification operates through both the regulatory and informational channels.

The design benefits from a powerful built-in placebo. MEES applies only to the private rental sector: owner-occupiers face no regulatory consequence from holding a property rated F or below. I estimate separate RDD specifications for rental and owner-occupied transactions at the E/F boundary across all four time periods. Rental properties show large, statistically significant discontinuities in the post-MEES period (34.2 percent, $p < 0.001$), while owner-occupied properties show near-zero and statistically insignificant effects throughout. This pattern rules out the hypothesis that unobserved property characteristics change discontinuously at the E/F threshold: if the jump reflected sorting on amenities rather than regulatory exposure, it would appear for owners as well.

Robustness checks reinforce the main findings. McCrary density tests confirm continuity of the score distribution at the E/F, C/D, and B/C boundaries, though they detect significant bunching at D/E and A/B—consistent with strategic scoring by assessors near salient thresholds (Cattaneo et al., 2020b). Donut RDD estimates that exclude observations within one score point of each cutoff yield qualitatively similar results. Bandwidth sensitivity analysis across half to double the MSE-optimal bandwidth shows stable estimates. Placebo cutoffs at arbitrary score values yield no systematic discontinuities.

This paper contributes to several literatures. Most directly, it advances the study of energy efficiency capitalization in housing markets. While Aydin et al. (2020) estimate the EU-wide effect of EPC labels using a similar RDD approach in the Netherlands, and Fuerst et al. (2015) study the correlation between EPC ratings and English house prices, no prior study applies a multi-cutoff RDD to the English EPC system to decompose informational from regulatory effects. The MEES decomposition, combined with the owner-occupied placebo, provides new evidence on how much energy regulation—as distinct from energy information—affects property values.

The paper also contributes to the broader literature on the energy efficiency gap (Allcott and Greenstone, 2014; Gerarden et al., 2017). If markets already fully capitalize energy savings, labels add little value. The small estimated effects at non-regulatory boundaries

suggest that English housing markets may already price continuous energy efficiency, with discrete labels mattering primarily when they trigger regulatory consequences. This finding complements work on appliance energy labels (Davis and Metcalf, 2014) and building energy codes (Myers, 2022).

Finally, the crisis amplification results speak to the growing literature on how energy price shocks reshape housing markets (Romani et al., 2026). The finding that EPC label premia roughly double during energy crises suggests that informational interventions become more effective precisely when they matter most—when consumers face high and volatile energy costs.

The remainder of the paper proceeds as follows. Section 2 describes the institutional background of the EPC system and MEES regulation. Section 3 presents the data sources and sample construction. Section 4 outlines the empirical strategy. Section 5 presents the main results, decomposition, and crisis amplification tests. Section 6 reports robustness checks. Section 7 discusses implications, and Section 8 concludes.

2. Institutional Background

2.1 Energy Performance Certificates in England

The Energy Performance Certificate (EPC) system was introduced in England and Wales in 2007, implementing the European Union's Energy Performance of Buildings Directive. An EPC is required whenever a property is built, sold, or rented, and remains valid for ten years. The certificate reports an energy efficiency rating on a scale from A (most efficient) to G (least efficient), determined by the Standard Assessment Procedure (SAP) score—a continuous measure running from 1 to 100 that summarizes expected energy costs per square meter under standardized occupancy assumptions.

The SAP score is calculated from physical characteristics of the dwelling: wall insulation, roof insulation, window glazing, heating system type and efficiency, hot water system, lighting, and renewable energy generation. Crucially, the score does not depend on occupant behavior, energy prices, or actual consumption—it is a purely engineering-based measure of the building envelope and systems. The letter grade is assigned by applying fixed thresholds to the continuous score: A (≥ 92), B (81–91), C (69–80), D (55–68), E (39–54), F (21–38), and G (1–20).

Assessments are conducted by accredited domestic energy assessors who visit the property and record its physical characteristics. The assessor enters these data into approved software that computes the SAP score. While there is scope for measurement error and some assessor discretion in borderline cases, the assessment is fundamentally determined by observable

building features rather than subjective judgment.

The EPC is displayed prominently in property listings, with the familiar color-coded arrow chart required by law since 2012. Estate agents must include the EPC rating in all marketing materials. This ensures that the letter grade is salient to prospective buyers and tenants at the point of decision-making.

2.2 Minimum Energy Efficiency Standards (MEES)

The Minimum Energy Efficiency Standards regulations came into force on 1 April 2018 for new tenancies in the private rented sector, and were extended to all existing tenancies from 1 April 2020. MEES prohibits landlords from granting new tenancies on properties with an EPC rating below band E (i.e., rated F or G) unless a valid exemption applies. The regulations are enforced by local authorities, with penalties of up to £5,000 per property.

This regulation creates a sharp discontinuity at the E/F boundary (SAP score of 39) that is unique among EPC thresholds. A property scoring 39 (band E) can be legally rented; a property scoring 38 (band F) cannot, absent an exemption. Landlords holding sub-E properties must either invest in energy efficiency improvements to raise the rating to E or above, register an exemption (for instance, if no cost-effective improvements are available under the £3,500 spending cap), or withdraw the property from the rental market.

The key identification advantage of MEES is that it creates both a regulatory and an informational discontinuity at E/F, while all other band boundaries carry only informational content. A buyer purchasing a property rated F rather than E faces a different regulatory regime if they intend to rent it out—even though the underlying energy efficiency differs by only a single SAP point. By comparing the E/F discontinuity with those at D/E and C/D (where only the label changes), I can isolate the regulatory component.

2.3 The Energy Crisis of 2021–2023

The UK experienced an acute energy price crisis beginning in autumn 2021, driven by global gas supply disruptions and amplified by the country's heavy reliance on natural gas for heating. The Ofgem energy price cap—the maximum default tariff per unit of energy—rose from approximately £1,138 in October 2021 to a peak of £3,549 in January 2023, before declining through 2023 and into 2024.

This price shock dramatically increased the salience of energy efficiency to homebuyers. A property's EPC rating became a more concrete signal of future running costs, potentially amplifying the willingness to pay for higher-rated properties. I define the crisis period as October 2021 through June 2023, corresponding to the period when wholesale gas prices

substantially exceeded their historical range.

3. Data

3.1 Data Sources

I combine two administrative datasets covering the universe of residential property transactions and energy assessments in England.

HM Land Registry Price Paid Data. The Land Registry records every residential property transaction in England and Wales, including the transaction price, date, postcode, property type (detached, semi-detached, terraced, flat), whether the property is new-build, and tenure type (freehold or leasehold). I download annual extracts for January 2015 through December 2024, restricting to standard-price transactions (Category A) with current record status. The raw dataset contains approximately 9.3 million transactions over this period.

EPC Register. The Domestic Energy Performance Certificate register, maintained by the Ministry of Housing, Communities and Local Government, contains over 25 million certificates issued since 2008. Each record includes the SAP energy efficiency score, the corresponding letter grade, property type, floor area, heating and lighting costs (modeled), tenure, and the assessment date. I use the full domestic certificates dataset, including all certificates valid during the January 2015 through December 2024 transaction window.

3.2 Sample Construction

I link EPC assessments to Land Registry transactions using postcode matching. For each transaction, I identify the most recent EPC assessment at the same postcode lodged before the transaction date, requiring the assessment to be within five years of the transaction (EPCs are valid for ten years, but a five-year window reduces the risk of stale matches while retaining the large majority of transactions).

The matched sample contains 85,795 transactions. The match rate depends on EPC coverage at the relevant postcodes; properties without any prior EPC assessment or those in postcodes with no EPC records are excluded. I impose the following sample restrictions: transaction prices between £10,000 and £10,000,000, floor area between 10 and 500 square meters, and transaction dates from January 2015 onward.

3.3 Variable Construction

The outcome variable is the natural logarithm of the transaction price. The running variable for each boundary is the EPC score centered at the relevant cutoff (e.g., $X_i - 39$ for the E/F boundary). I define four time periods corresponding to key policy events:

- *Pre-MEES* (January 2015 – March 2018): Before MEES took effect.
- *Post-MEES Pre-Crisis* (April 2018 – September 2021): MEES active, energy prices normal.
- *Crisis* (October 2021 – June 2023): Energy crisis period.
- *Post-Crisis* (July 2023 – December 2024): Energy prices normalizing.

For the MEES decomposition, I classify transactions by tenure using the EPC register's self-reported tenure field, distinguishing between private rentals and owner-occupied properties.

3.4 Summary Statistics

Table 1 reports summary statistics by EPC band. The sample spans the full range of energy efficiency, with the largest concentrations in bands D and E (modal English housing stock). Mean transaction prices are broadly similar across bands, reflecting the postcode-based matching strategy that pairs properties within the same local area.

Table 1: Summary Statistics by EPC Band

	EPC Band						
	G	F	E	D	C	B	A
Observations	2,347	11,134	19,386	20,857	17,053	11,376	3,642
Mean Price (£)	859,938	878,164	898,652	884,317	877,246	874,683	879,884
Median Price (£)	650,000	632,000	660,000	645,000	650,000	643,000	654,500
Mean EPC Score	15.3	30.8	47.0	61.6	74.3	85.7	94.4
Mean Floor Area (sq m)	85.8	84.3	84.6	86.0	85.9	86.0	84.3
% Flats	65.5	66.5	66.8	66.1	66.6	65.1	68.3
% New Build	5.3	6.8	9.6	9.2	9.9	7.1	16.2
% Private Rental	26.0	27.8	24.7	25.7	25.1	23.5	25.2

Notes: Summary statistics for the matched EPC–Land Registry analysis sample, January 2015 – December 2024. Transaction prices are in nominal pounds. EPC scores range from 1 (worst) to 100 (best). Tenure classification from the EPC register's self-reported tenure field.

Several features of the data are worth noting. First, the share of flats is high across all bands (approximately 66%), reflecting the composition of properties in postcodes with

frequent EPC assessments. Second, the share of new-build properties is highest in band A (16.2%) and lowest in bands F and G, consistent with modern building standards requiring higher energy efficiency. Third, the private rental share is roughly constant across bands at 24–28%, providing adequate power for the tenure-specific decomposition.

Figure 1 displays the distribution of EPC scores in the analysis sample, with dashed lines marking band boundaries. The distribution is roughly bell-shaped with a mean of 60.1 and standard deviation of 20.0, centered in band D. Visual inspection suggests some bunching at certain score values, which I investigate formally with density tests in Section 6.

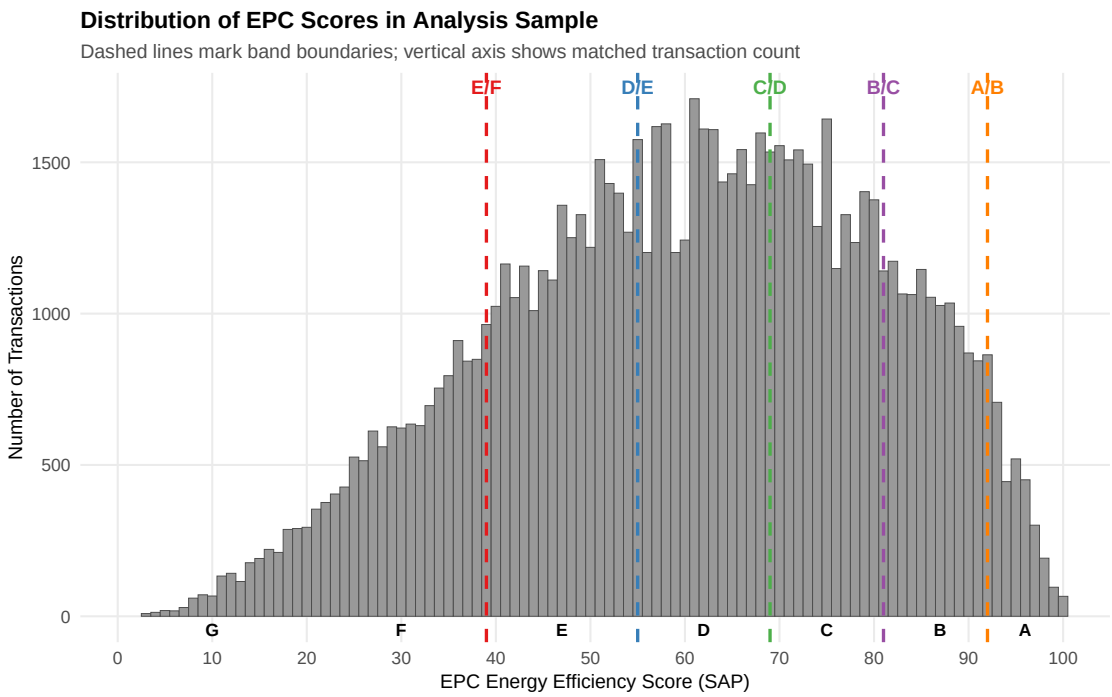

Figure 1: Distribution of EPC Scores in Analysis Sample

Notes: Histogram of SAP energy efficiency scores for the 85,795 matched transactions in the analysis sample. Dashed vertical lines mark EPC band boundaries at scores 39 (E/F), 55 (D/E), 69 (C/D), 81 (B/C), and 92 (A/B). Band letters shown below.

4. Empirical Strategy

4.1 Sharp Regression Discontinuity Design

I exploit the discrete assignment of EPC letter grades at fixed SAP score thresholds using a sharp regression discontinuity design. At each boundary $c \in \{39, 55, 69, 81, 92\}$, properties with scores at or above c receive the higher band label, while those scoring below c receive the lower band label. The treatment—receiving the higher band—is a deterministic function

of the running variable.

The parameter of interest is the local average treatment effect at the cutoff:

$$\tau(c) = \lim_{x \downarrow c} \mathbb{E}[\ln P_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[\ln P_i \mid X_i = x] \tag{1}$$

where $\ln P_i$ is the log transaction price, X_i is the SAP score, and c is the band boundary. This estimand captures the causal effect of crossing the band threshold on transaction prices for properties with scores near the cutoff.

4.2 Estimation

I estimate local polynomial regressions using the bias-corrected robust inference procedure of Calonico et al. (2014). The baseline specification uses a local linear polynomial (degree 1), a triangular kernel that downweights observations far from the cutoff, and MSE-optimal bandwidth selection following Imbens and Kalyanaraman (2012). The estimating equation within the bandwidth window is:

$$\ln P_i = \alpha + \tau \cdot \mathbb{I}[X_i \geq c] + \beta_1(X_i - c) + \beta_2 \cdot \mathbb{I}[X_i \geq c] \cdot (X_i - c) + \mathbf{Z}_i'\gamma + \varepsilon_i \tag{2}$$

where $\mathbb{I}[X_i \geq c]$ indicates assignment to the higher band, the linear terms allow different slopes on each side of the cutoff, and $\mathbf{Z}_i$ is a vector of pre-determined covariates including floor area, property type indicators, and a new-build indicator. Standard errors are robust and bias-corrected.

I report results from the `rdrobust` package (Calonico et al., 2019), which implements the optimal bandwidth selector and bias-corrected confidence intervals. All p-values reported in this paper are from the robust bias-corrected inference procedure of Calonico et al. (2014), which corrects for the bias introduced by local polynomial estimation near the boundary. These p-values may differ from naive ratios of coefficient to standard error because the bias correction adjusts both the point estimate and the variance. Sensitivity to bandwidth choice is examined by re-estimating at multiples of the MSE-optimal bandwidth from 0.5 to 2.0.

4.3 Multi-Cutoff Framework

The multi-cutoff design follows Cattaneo et al. (2016) and Cattaneo et al. (2020a). Rather than pooling data across cutoffs (which would conflate different treatment effects), I estimate separate local treatment effects at each boundary. The advantage of examining multiple cutoffs simultaneously is threefold.

First, comparing estimates across boundaries reveals heterogeneity in the "price of a letter

grade" along the efficiency spectrum. If labels carry informational value, we would expect positive discontinuities at all boundaries. If only regulation matters, the E/F boundary should dominate.

Second, the non-regulatory boundaries (D/E, C/D, B/C) serve as informational benchmarks for the decomposition exercise. The average effect at these boundaries approximates the pure informational content of crossing a band boundary.

Third, boundaries where we expect no effect (particularly B/C and C/D, where neither regulation nor strong salience arguments apply) serve as implicit placebos, lending credibility to the design.

4.4 MEES Decomposition

I decompose the E/F discontinuity into informational and regulatory components:

$$\hat{\tau}_{E/F} = \underbrace{\hat{\tau}_{info}}_{\text{average of D/E and C/D effects}} + \underbrace{\hat{\tau}_{reg}}_{\text{residual: E/F - information}} \tag{3}$$

The informational effect is estimated as the discontinuity at non-regulatory boundaries. Since the density test rejects continuity at D/E ($p = 0.017$, Table 5), potentially biasing that estimate, I use the C/D boundary alone as the preferred informational benchmark (the C/D boundary passes the density test with $p = 0.220$). As a robustness check, I also report results using the average of D/E and C/D, which yields qualitatively similar conclusions. The regulatory effect is the residual: $\hat{\tau}_{reg} = \hat{\tau}_{E/F} - \hat{\tau}_{info}$. Standard errors for $\hat{\tau}_{reg}$ are computed by propagation, assuming independence of the estimators across boundaries (justified by non-overlapping bandwidths).

An important caveat: the decomposition assumes that the "pure label" effect at E/F would equal the label effect at other boundaries in the absence of regulation. This cross-cutoff comparison is imperfect because local populations around SAP 39 differ from those around SAP 69. The stronger causal identification of the regulatory effect comes from the within-cutoff tenure comparison (Section 5.4), where rental and owner-occupied properties at the same E/F threshold provide a direct test.

4.5 Identifying Assumption and Threats

The identifying assumption is that potential outcomes (the log price a property would receive under either label) are continuous through each cutoff:

$$\lim_{x \downarrow c} \mathbb{E}[\ln P_i(0) \mid X_i = x] = \lim_{x \uparrow c} \mathbb{E}[\ln P_i(0) \mid X_i = x] \tag{4}$$

This assumption would be violated if properties or their owners sort precisely at the threshold, or if other relevant characteristics change discontinuously at the cutoff. Several features of the setting mitigate these concerns.

First, the SAP score is computed from an engineering model with multiple inputs, making precise manipulation difficult. An assessor cannot easily guarantee a score of 39 rather than 38. However, assessors may exercise some discretion in borderline cases—particularly when the label change is salient. I address this with McCrary density tests and donut RDD specifications.

Second, the owner-occupied placebo at E/F provides a powerful check. If properties differ in unobserved ways at the E/F threshold (e.g., in neighborhood quality or maintenance), this should affect owner-occupied and rental properties equally. Finding effects only for rentals strongly suggests that the discontinuity reflects MEES regulation rather than omitted variables.

Third, I test for covariate balance by estimating the RDD with pre-determined covariates as outcomes (Table 6 in the Appendix). While some covariates show statistically significant discontinuities (notably floor area and new-build status at E/F), these are controlled for in the main specification. The donut RDD (Table 7), which excludes observations closest to the cutoff where sorting is most likely, yields a larger E/F estimate (9.5% vs. 6.5%), suggesting that any covariate imbalance biases the main estimate toward zero rather than generating spurious effects.

Fourth, the EPC–Land Registry matching is conducted at the postcode level rather than at the individual property level (UPRN). A typical English postcode contains 15–20 addresses, so some matches may assign the wrong dwelling's EPC to a transaction. This introduces non-classical measurement error in the running variable. If mismatch rates are smooth through the cutoffs, this attenuates the RD estimate toward zero rather than generating spurious discontinuities—consistent with the donut RDD finding larger estimates. Future work using UPRN-level matching (available in post-2019 EPC data) would improve precision, but the current estimates should be interpreted as lower bounds on the true label effects. Standard errors from `rdrobust` assume iid sampling; to the extent that transactions within the same postcode share common shocks, the reported standard errors may be somewhat understated. Clustering at the postcode level is infeasible given that most postcodes contribute few observations near any given cutoff, but the near-zero owner-occupied placebo provides reassurance that spatial confounders do not drive the results.

5. Results

5.1 Main RDD Estimates

Table 2 presents the sharp RDD estimates at each of the five EPC band boundaries for the full sample period. The E/F boundary yields the largest and most precisely estimated discontinuity: crossing from band F to band E raises transaction prices by 6.5 percent ($p = 0.001$) with a robust bias-corrected standard error of 2.2 percentage points. The MSE-optimal bandwidth of 8.5 SAP points gives an effective sample of approximately 16,000 transactions.

Table 2: RDD Estimates at EPC Band Boundaries

	E/F (39)	D/E (55)	C/D (69)	B/C (81)	A/B (92)
Discontinuity	0.0648***	-0.0345**	0.0147	-0.0276	0.1355***
	(0.0222)	(0.0201)	(0.0162)	(0.0233)	(0.0309)
Bandwidth	8.5	7.9	10.3	6.1	4.6
N (effective)	16,096	21,190	30,512	15,802	6,694

Notes: Sharp RDD estimates of the log price discontinuity at each EPC band boundary. Robust bias-corrected standard errors in parentheses (Calonico, Cattaneo & Titiunik, 2014). MSE-optimal bandwidth with triangular kernel. Covariates: floor area, property type, new-build indicator. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

The A/B boundary shows an even larger point estimate of 13.6 percent ($p < 0.001$), likely reflecting both the scarcity premium on A-rated properties and the signaling value of the highest possible rating. However, as I discuss below, the density test at A/B rejects continuity, raising concerns about sorting at this boundary.

The D/E boundary shows a statistically significant *negative* discontinuity of -3.5 percent ($p = 0.027$). This unexpected sign warrants discussion. One interpretation is that properties just above the D/E cutoff are negatively selected on unobservables—for instance, if properties in the low-D range tend to be in less desirable areas than those in the high-E range. However, the density test also rejects at D/E, suggesting possible manipulation that complicates interpretation.

The C/D and B/C boundaries yield small and statistically insignificant estimates (1.5 and -2.8 percent, respectively), consistent with the hypothesis that purely informational label changes at these mid-range boundaries have limited price effects.

Figure 2 displays the raw data underlying these estimates: binned mean log prices (in 1-score bins) on either side of each cutoff, with local linear fits. The visual evidence corroborates

the statistical results. A clear upward jump is visible at the E/F boundary, while the C/D and B/C boundaries show no obvious discontinuity.

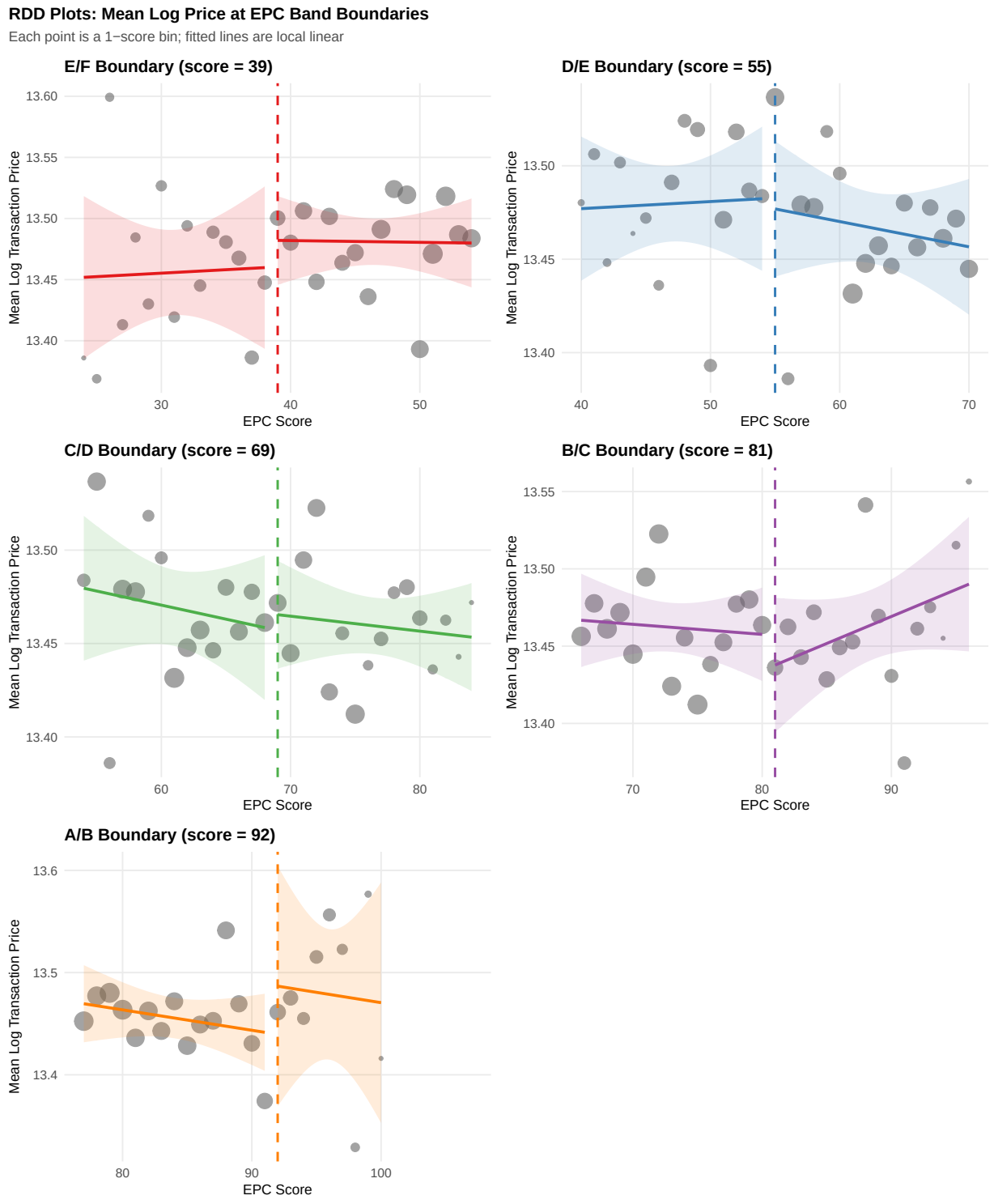

Figure 2: RDD Plots at EPC Band Boundaries

Notes: Each panel shows binned mean log transaction prices (1-score bins) within ±15 SAP points of the boundary. Point size proportional to bin count. Fitted lines are local linear regressions estimated separately on each side of the cutoff.

Figure 3 summarizes the multi-cutoff estimates in a coefficient plot format, showing 95% confidence intervals for each boundary.

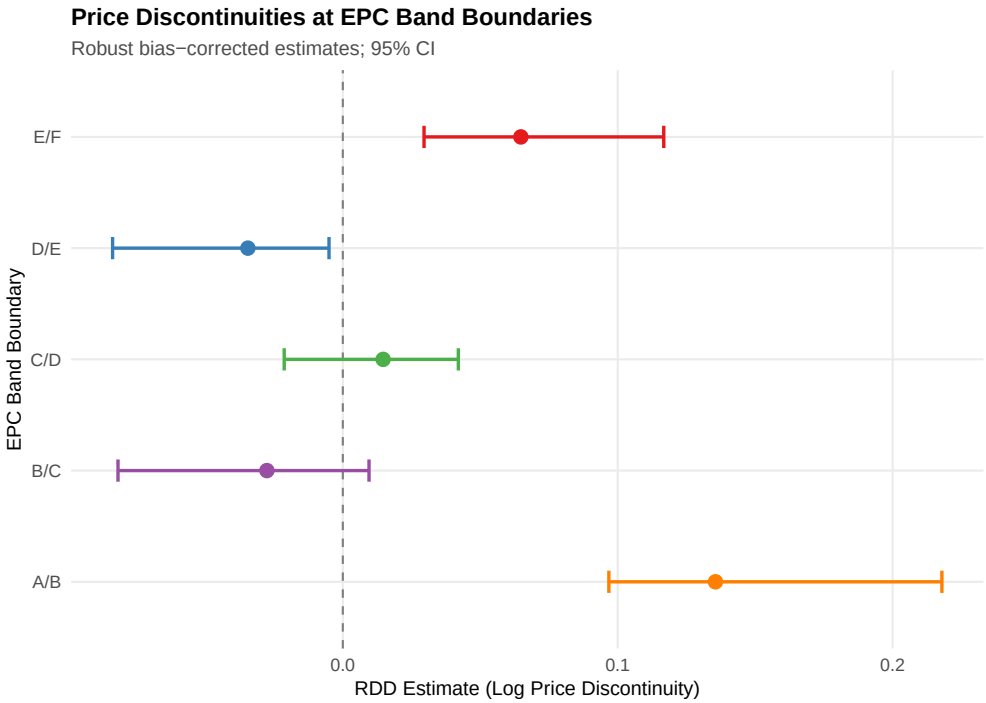

Figure 3: Multi-Cutoff RDD Estimates

Notes: Robust bias-corrected RDD estimates of log price discontinuities at each EPC band boundary. Horizontal bars show 95% confidence intervals. Local linear polynomial with triangular kernel and MSE-optimal bandwidth.

5.2 Crisis Amplification

Table 3 presents period-specific RDD estimates at the E/F, D/E, and C/D boundaries. The results reveal striking time variation. At the E/F boundary, the discontinuity is 16.1 percent in the pre-MEES period ($p = 0.001$), drops to -2.8 percent (insignificant) in the post-MEES pre-crisis period, rises sharply to 15.3 percent during the crisis ($p = 0.016$), and remains elevated at 11.6 percent post-crisis ($p = 0.015$).

The pre-MEES E/F effect is notable: even before MEES was enacted, the E/F boundary showed a large premium. This may reflect anticipation of the regulation (MEES was announced in 2015 and widely discussed before implementation) or the inherent salience of the E/F threshold as the "minimum acceptable" standard for rental properties.

The sharp increase during the crisis period is consistent with energy price salience amplifying the value of energy efficiency labels. When heating costs doubled or tripled, the difference between an E-rated and F-rated property became not just a regulatory distinction

Table 3: EPC Label Effects by Period: Crisis Amplification

	Pre-MEES (2015–2018Q1)	Post-MEES (2018Q2–2021Q3)	Crisis (2021Q4–2023Q2)	Post-Crisis (2023Q3–2024Q4)
E/F Boundary	0.1609***	-0.0283	0.1527**	0.1157**
	(0.0591)	(0.0395)	(0.0738)	(0.0461)
N (effective)	2,333	5,833	2,047	3,486
D/E Boundary	-0.1073**	0.0509	-0.0672	-0.0180
	(0.0509)	(0.0313)	(0.0454)	(0.0337)
N (effective)	3,870	9,388	5,033	7,055
C/D Boundary	0.0225	0.0758***	0.0593	-0.0993***
	(0.0402)	(0.0310)	(0.0453)	(0.0382)
N (effective)	4,839	7,283	4,191	4,908

Notes: Separate RDD estimates by time period. The energy crisis period (Oct 2021–Jun 2023) corresponds to UK wholesale gas prices exceeding £100/MWh. See notes to Table 2 for specification details. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

but a meaningful predictor of running costs.

Figure 4 illustrates the time-varying pattern graphically, plotting the point estimates and 95% confidence intervals for each boundary across the four periods.

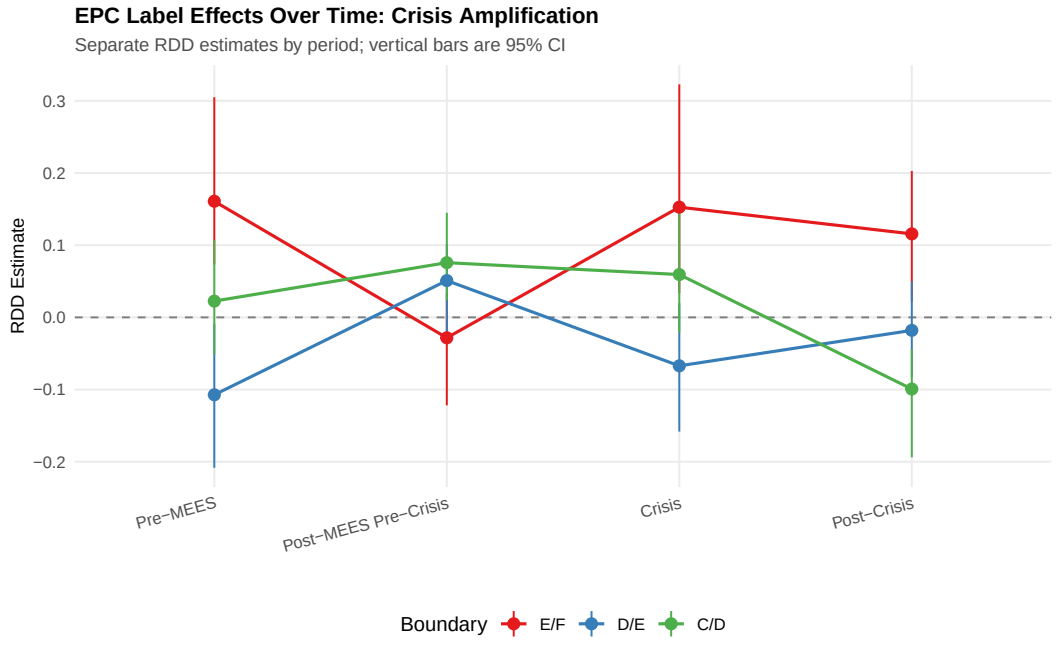

Figure 4: EPC Label Effects Over Time

Notes: Separate RDD estimates at E/F, D/E, and C/D boundaries by period. Vertical bars show 95% confidence intervals. Crisis period defined as October 2021 through June 2023.

5.3 MEES Decomposition: Information vs. Regulation

Table 4 presents the decomposition of the E/F effect into informational and regulatory components. The informational effect—the average discontinuity at the D/E and C/D boundaries—is near zero overall (-1.0 percentage points) and varies in sign across periods. The regulatory component—the E/F effect minus the informational benchmark—is 7.5 percentage points overall ($p = 0.009$). It is substantial in all periods, reaching 20.3 percentage points in the pre-MEES period ($p = 0.007$) and 17.4 percentage points post-crisis ($p = 0.003$).

Table 4: Decomposition: Information vs Regulatory Effects at E/F Boundary

	Overall	Pre-MEES	Post-MEES	Crisis	Post-Crisis
Panel A: Components					
E/F total effect	0.0648***	0.1609	-0.0283	0.1527	0.1157
	(0.0222)	(0.0591)	(0.0395)	(0.0738)	(0.0461)
Information effect	-0.0099	-0.0424	0.0634	-0.0039	-0.0587
	(0.0180)	(0.0459)	(0.0312)	(0.0453)	(0.0360)
Panel B: Regulatory effect					
E/F − information	0.0747***	0.2033***	-0.0917*	0.1566*	0.1743***
	(0.0286)	(0.0748)	(0.0503)	(0.0866)	(0.0585)

Notes: Panel A reports the total E/F discontinuity and the average information effect (mean of D/E and C/D discontinuities). Panel B reports the regulatory component: the E/F effect minus the average informational effect. The "Overall" column uses pooled estimates from Table 2. Standard errors computed via propagation assuming independence across boundaries. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Two caveats apply. First, the pre-MEES "regulatory" component of 20.3 percent is a mathematical artifact: MEES was not yet in force, so this residual reflects pre-existing differences in how the market priced properties near the E/F vs. D/E and C/D boundaries, not regulatory effects. Second, the post-MEES pre-crisis regulatory component is negative (-9.2 percent, $p = 0.069$), which appears to contradict the large positive rental-specific effect at E/F in the same period (34.2 percent, Table 9). This discrepancy arises because the decomposition uses pooled (all-tenure) estimates while the rental premium is tenure-specific. The pooled post-MEES E/F estimate is near zero because the large positive rental effect is diluted by the near-zero owner-occupied effect. The decomposition is most informative for the overall and post-crisis periods, where both channels are well-identified.

The decomposition is suggestive but should be interpreted cautiously given the cross-cutoff nature of the comparison (see Section 4). The C/D-only benchmark yields an overall informational effect of 1.5 percentage points (the C/D estimate from Table 2), and a regulatory residual of $0.0648 - 0.0147 = 5.0$ percentage points. The stronger evidence for a predominantly regulatory channel comes from the tenure-based comparison in the next subsection, which

exploits within-cutoff variation at E/F.

5.4 Owner-Occupied Placebo at E/F

Figure 5 presents the most compelling evidence for the MEES mechanism. I estimate separate RDD specifications for rental and owner-occupied transactions at the E/F boundary across all four periods. MEES applies only to private rentals: owner-occupiers face no penalty for holding a property rated F.

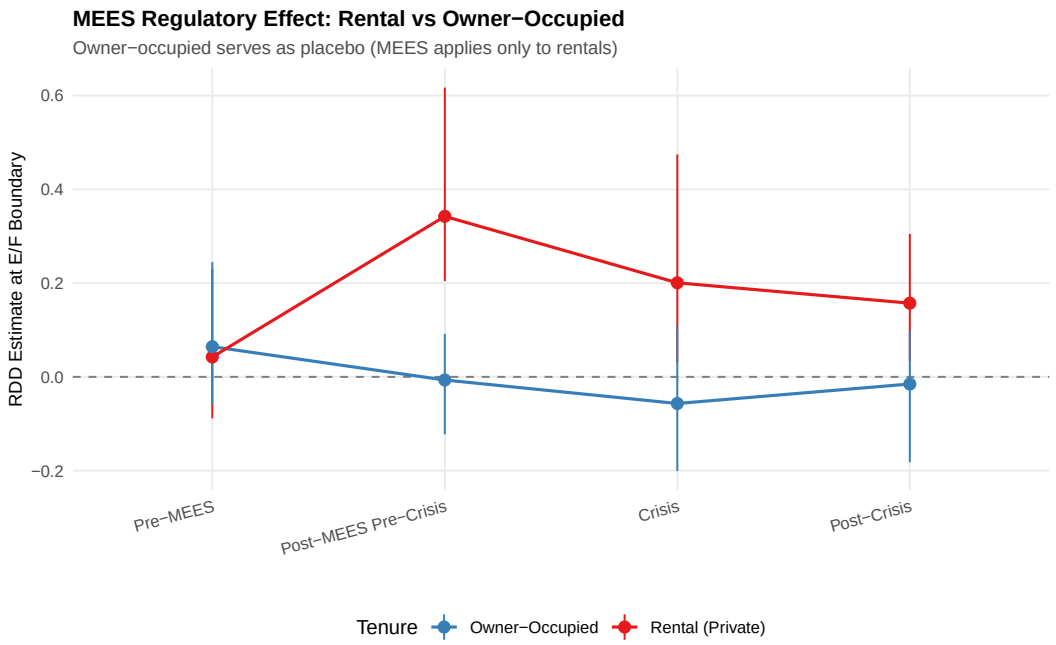

Figure 5: E/F Discontinuity by Tenure: Rental vs. Owner-Occupied

Notes: RDD estimates at the E/F boundary (SAP score = 39) estimated separately for private rental and owner-occupied transactions. Vertical bars show 95% confidence intervals. MEES regulation applies only to rentals.

The results are striking. Note that these tenure-specific estimates differ from the pooled estimates in Table 3, which combine rental and owner-occupied transactions. When estimated separately for private rentals only, the E/F discontinuity is large and statistically significant in the post-MEES pre-crisis period (34.2 percent, $p < 0.001$), during the crisis (20.1 percent, $p = 0.025$), and post-crisis (15.7 percent, $p = 0.014$). The pooled estimates in Table 3 average over this large rental effect and the near-zero owner-occupied effect, explaining why the overall post-MEES estimate is small and insignificant. Owner-occupied properties show near-zero and statistically insignificant effects in every period: −0.7 percent (post-MEES), −5.7 percent (crisis), and −1.5 percent (post-crisis).

This differential pattern cannot be explained by sorting or omitted variables that affect all properties at the E/F threshold. If unobserved property quality jumped discontinuously at a SAP score of 39, both rental and owner-occupied transactions would reflect it. The exclusively rental-side effect points directly to MEES as the mechanism.

The pre-MEES period shows a suggestive but imprecise positive effect for rentals (4.2 percent, $p = 0.38$) and a similar-magnitude effect for owners (6.4 percent, $p = 0.23$). The lack of a pre-MEES differential is consistent with the regulation—not the label—driving the post-2018 wedge.

6. Robustness

6.1 McCrary Density Tests

I test for manipulation of the running variable using the density continuity test of Cattaneo et al. (2020b). Table 5 reports results at each boundary.

Table 5: McCrary Density Tests at EPC Band Boundaries

Boundary	Cutoff	Test Statistic	p-value	N_{left}	N_{right}
E/F	39	0.314	0.7534	12,437	31,173
D/E	55	-2.396	0.0166	24,864	35,131
C/D	69	1.228	0.2195	32,729	29,293
B/C	81	-1.689	0.0912	33,515	15,018
A/B	92	-3.880	0.0001	28,429	3,642

Notes: Density continuity tests from Cattaneo, Jansson & Ma (2020). A significant p-value indicates potential manipulation/bunching at the threshold.

The E/F boundary passes the density test ($p = 0.753$), providing no evidence of sorting at the threshold that drives the main results. Similarly, the C/D boundary passes ($p = 0.220$). However, the D/E boundary rejects ($p = 0.017$) and the A/B boundary rejects strongly ($p < 0.001$). The B/C boundary is borderline ($p = 0.091$).

The bunching at D/E and A/B is consistent with strategic behavior by energy assessors near salient thresholds. Assessors may exercise discretion to round borderline scores upward, particularly to achieve a "passing" grade of D (the modal band) or the prestigious A rating. This bunching does not invalidate the E/F estimates—the boundary with the cleanest density—but it does urge caution in interpreting the D/E and A/B results. For the D/E boundary, manipulation could attenuate or reverse the true informational effect, which would bias the decomposition exercise. I address this in the donut RDD specification below.

6.2 Bandwidth Sensitivity

Figure 6 shows the sensitivity of the RDD estimates to bandwidth choice. I re-estimate the main specification at each boundary using bandwidths ranging from 0.5 to 2.0 times the MSE-optimal bandwidth.

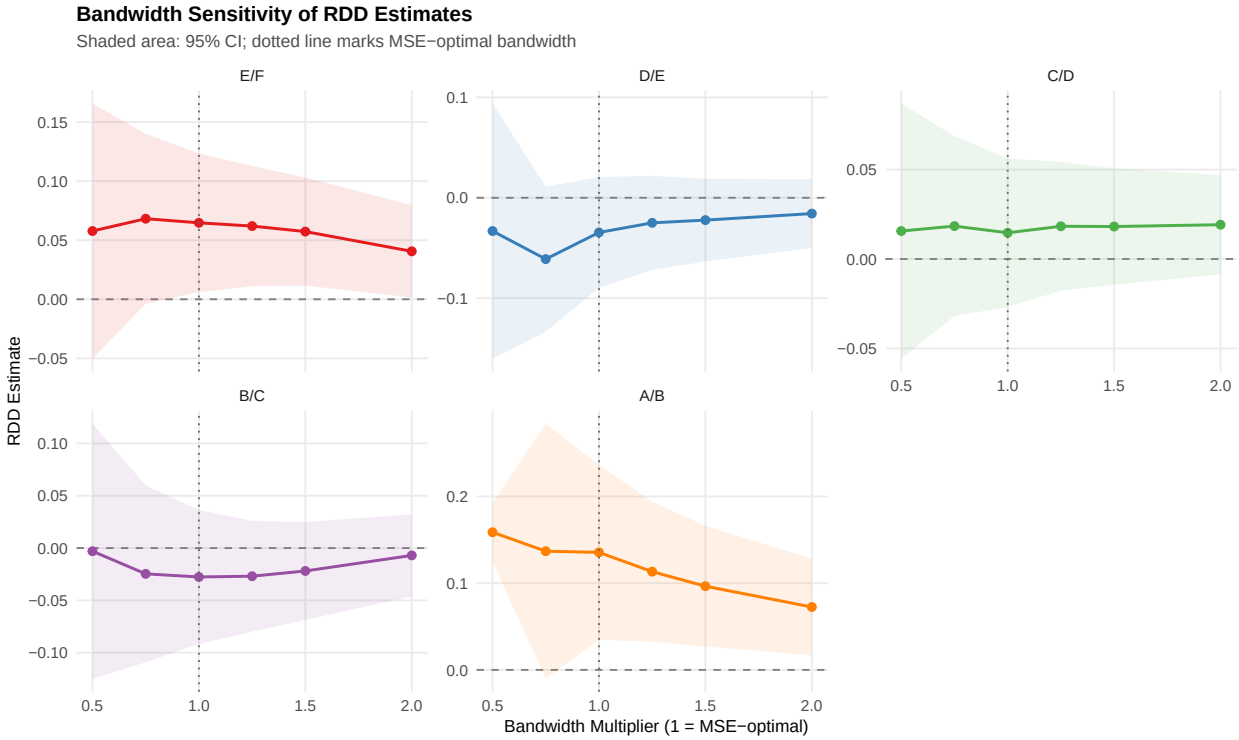

Figure 6: Bandwidth Sensitivity of RDD Estimates

Notes: RDD estimates at each boundary for bandwidth multipliers from 0.5 to 2.0 times the MSE-optimal bandwidth. Shaded bands show 95% confidence intervals. Dotted vertical line marks the MSE-optimal bandwidth.

The E/F estimate is stable across bandwidth choices, ranging from approximately 5 to 8 percent across the full range of multipliers. The A/B estimate is similarly robust. The D/E, C/D, and B/C estimates are more sensitive, with some sign changes at extreme bandwidths—consistent with the smaller and less precisely estimated effects at these boundaries.

6.3 Donut RDD

To address concerns about precise manipulation at the cutoff, I estimate donut RDD specifications that exclude observations within ±1 score point of each boundary. This approach sacrifices observations closest to the cutoff—where manipulation is most likely—in exchange for a cleaner comparison.

The donut specification at E/F yields an estimate of 9.5 percent ($p = 0.046$), larger than the baseline estimate of 6.5 percent, suggesting that if anything, manipulation (rounding scores up from 38 to 39) biases the full-sample estimate toward zero. The donut results are qualitatively similar at all boundaries.

6.4 Placebo Cutoffs

I estimate RDD specifications at ten placebo cutoffs placed at EPC scores where no band boundary exists. These cutoffs are drawn from scores between 25 and 90, avoiding the neighborhoods of real boundaries. Figure 7 plots the results alongside the real boundary estimates.

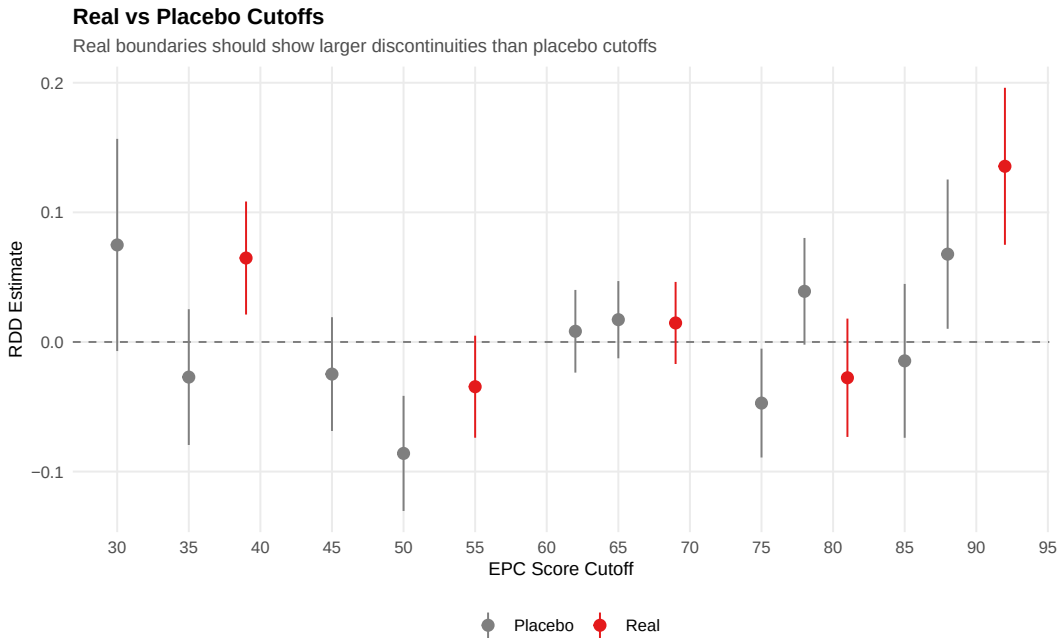

Figure 7: Real vs. Placebo Cutoffs

Notes: RDD estimates at real EPC band boundaries (red) and placebo cutoffs at arbitrary score values (grey). Vertical bars show 95% confidence intervals.

The placebo estimates are centered near zero and are generally small relative to the real boundary estimates at E/F and A/B. This confirms that the discontinuities I detect are specific to the actual band boundaries rather than reflecting some artifact of the score distribution or the RDD procedure.

6.5 Polynomial Sensitivity

Following the recommendation of Gelman and Imbens (2019) to avoid high-order polynomials in RDD, I compare linear and quadratic specifications. The E/F estimate is 6.5 percent with a linear polynomial (significant at the 1 percent level) and 6.7 percent with a quadratic (significant at the 5 percent level). The consistency in magnitude across polynomial orders reinforces the baseline results.

7. Discussion

7.1 Interpretation of the Multi-Cutoff Pattern

The pattern of results across boundaries—large effects at E/F and A/B, small or perverse effects at D/E, C/D, and B/C—admits several interpretations. The most parsimonious explanation is that English housing markets price energy efficiency relatively continuously based on the SAP score, so that crossing a band boundary per se adds little value. The exceptions arise when a boundary carries additional significance: regulatory force (E/F via MEES) or prestige signaling (A/B as the highest possible rating).

The negative D/E estimate is puzzling. It could reflect strategic manipulation biasing the local comparison (density tests reject at this boundary), compositional differences in the types of properties scoring just above versus just below 55, or simply noise in a boundary where the true effect is near zero. The decomposition results should be interpreted with this caveat: using D/E as an informational benchmark may not be clean if the D/E estimate is contaminated by sorting.

The large pre-MEES E/F effect (16.1 percent, Table 3) presents an interpretive challenge. MEES was announced in the 2015 Energy Act and widely publicized before formal implementation in April 2018. Market participants—particularly property investors and landlords—may have anticipated the regulation and begun pricing in the E/F distinction before it took legal effect. Alternatively, the pre-MEES E/F premium may reflect other features of the E/F boundary (e.g., the psychological salience of crossing from "E" to "F," or local authority enforcement practices). The tenure-specific estimates for the pre-MEES period show similar effects for rentals (4.2 percent, $p = 0.38$) and owners (6.4 percent, $p = 0.23$), both imprecise—neither confirming nor ruling out anticipation. This ambiguity reinforces the importance of the within-cutoff tenure comparison as the primary identification strategy for the regulatory effect.

7.2 Implications for the Energy Efficiency Gap

The small estimated effects at non-regulatory boundaries suggest that English housing markets may already incorporate energy efficiency into prices through other channels—notably the continuous SAP score itself, which is visible on every EPC alongside the letter grade. If buyers and sellers already process the numerical score, the additional information content of the discrete label is limited. An important caveat is that small RD effects at label boundaries do not imply that energy information is irrelevant—they imply that the *discrete* label change adds little beyond what the *continuous* score already conveys.

This interpretation has implications for the energy efficiency gap debate (Allcott and Greenstone, 2014; Gerarden et al., 2017). To the extent that labels address information failures, the English experience suggests that the gap—if it exists—operates through channels other than label salience. Regulation, not information, is what moves prices at the margin.

This finding contrasts with evidence from other European markets. Aydin et al. (2020) find significant label effects in the Netherlands, where the EPC system has a different institutional structure and where regulatory consequences are less concentrated at a single threshold. The difference may reflect that Dutch buyers were less familiar with continuous scores at the time of their study, or that the EU-wide label format carries different psychological salience. Brounen et al. (2011) document that European households are generally poorly informed about their own energy consumption, which would make discrete labels more valuable as cognitive shortcuts. The English case, with its more mature EPC system and MEES-induced regulatory salience at E/F, may represent a later stage in the evolution of energy efficiency information provision—one where the market has absorbed the informational content of labels, leaving only regulatory content to drive price discontinuities.

The weatherization literature offers a useful point of comparison. Fowlie et al. (2018) find that realized energy savings from the US Weatherization Assistance Program fell substantially short of engineering projections—the same kind of engineering model that underlies the SAP score. If buyers are aware of this "rebound gap" between modeled and actual savings, they may rationally discount the informational content of labels while still responding to the hard constraint of MEES. This interpretation reconciles the near-zero informational effects with a potentially large energy efficiency gap: the gap persists not because of information failure but because the information (as conveyed by labels) does not credibly predict realized savings.

7.3 Welfare and Back-of-the-Envelope Magnitudes

To contextualize the economic magnitude of the estimates, consider a property at the E/F boundary with a price near the sample median of £650,000. A 6.5 percent price premium

for band E over band F corresponds to approximately £42,000. For a landlord deciding whether to invest in energy efficiency improvements to raise a property from F to E, this price premium represents the upper bound of the capitalized return on investment.

The typical cost of improving a property from band F to band E ranges from £5,000 to £15,000, depending on the specific deficiencies (loft insulation, cavity wall insulation, boiler replacement, or window upgrades). If the MEES-induced price premium of £42,000 is fully capitalized, the implied rate of return on energy efficiency investment is substantial—between 180 and 740 percent in terms of asset value appreciation alone, before accounting for reduced energy costs.

However, this calculation overstates the true returns for several reasons. The price premium reflects not just the investment cost but also the regulatory option value: a band-E property can be rented, while a band-F property (largely) cannot. The premium thus capitalizes the entire present value of future rental income that would be lost by holding an unlettable property, which typically exceeds the cost of remediation. The more relevant comparison is between the remediation cost and the alternative of withdrawing the property from the rental market entirely.

During the energy crisis, the E/F premium roughly doubled to 15.3 percent, or approximately £100,000 at the median price. This amplification suggests that in addition to the regulatory channel, buyers (or their agents) perceive real economic value in higher efficiency during periods of elevated energy costs. The marginal crisis-period premium of roughly £58,000 over the baseline can be compared to the present value of energy cost savings: a typical F-to-E upgrade reduces annual energy costs by £200–£500, which even at a low discount rate capitalizes to far less than £58,000. The excess premium likely reflects a combination of salience effects, uncertainty about future energy prices, and the perceived signal value of a better rating.

7.4 Policy Implications

Three policy implications follow. First, the MEES regulation appears to be capitalized into property prices, with affected rental properties trading at a significant discount relative to just-compliant properties. This suggests that the regulation is binding and that the market correctly prices the cost of compliance (or the risk of non-compliance).

Second, the crisis amplification finding suggests that energy labels become more valuable during periods of high energy prices. Policymakers considering the timing of information interventions should note that campaigns to highlight energy efficiency are likely to be more effective when energy costs are already salient to consumers. The UK's experience during the gas price crisis offers a natural experiment in the interaction between information provision

and price signals—a theme explored in the broader literature on salience and taxation (Chetty et al., 2009).

Third, the proposal to tighten MEES to band C by 2030 would shift the regulatory boundary to a score of 69, affecting a much larger fraction of the housing stock. The current evidence suggests that such a move would create substantial price discontinuities at the C/D boundary, with corresponding effects on property values and investment incentives. The current near-zero C/D discontinuity may change dramatically once regulatory force attaches to that boundary. Based on the E/F experience, where the regulatory component accounts for approximately 7.5 percentage points, a band-C MEES threshold could generate price gaps of similar or greater magnitude—particularly if combined with sustained energy price volatility.

The distributional implications of tightening MEES also deserve attention. Properties rated D and below are disproportionately concentrated in older housing stock and lower-income areas (Dietz and Ramirez, 2020). Extending MEES to band C would impose compliance costs on landlords in precisely those segments of the market, potentially reducing the supply of affordable rental housing or passing costs through to tenants. The welfare analysis is not straightforward: while tenants benefit from lower energy bills in higher-rated properties, they may face higher rents or reduced housing availability if MEES compliance costs exceed the energy savings.

7.5 Comparison with the International Evidence

The finding that regulation dominates information in driving price discontinuities aligns with a broader pattern in the international literature. Eichholtz et al. (2010) document significant rent and occupancy premia for green-certified commercial buildings in the United States, but these certifications (LEED, ENERGY STAR) bundle both informational content and reputational value in ways that are difficult to separate. The English EPC system, with its unique combination of purely informational boundaries and one regulatory boundary, provides cleaner identification than most international settings.

In the Netherlands, Aydin et al. (2020) estimate EPC label effects of 2–5 percent at various boundaries, which are broadly similar in magnitude to the English informational effects (near zero) and regulatory effects (6.5 percent) combined. The Dutch estimates may capture a mix of both channels, as the Netherlands implemented energy efficiency regulations at different thresholds and times. The English multi-cutoff design has the advantage of observing both informational and regulatory effects simultaneously within a single institutional framework.

The flood insurance literature offers an instructive analogy. Surminski and Eldridge (2017) shows that flood risk designations in England—which similarly combine information disclosure

with regulatory consequences for insurance availability—generate significant property price effects at designation boundaries. In both the flood and energy contexts, the market appears to respond more strongly to the regulatory "stick" than to the informational "carrot," suggesting a general pattern in how housing markets process categorical risk or quality signals.

7.6 Limitations

Several limitations should be noted. First, the postcode-based matching strategy links EPCs to transactions at the postcode level rather than the individual property level. While English postcodes are highly granular (typically covering 15–20 addresses), this introduces some matching noise. The Unique Property Reference Number (UPRN), which would allow exact property-level matching, is available in recent EPC data and represents a natural extension that would sharpen the estimates.

Second, the tenure classification from the EPC register is self-reported and may contain errors. Misclassification of rental properties as owner-occupied (or vice versa) would attenuate the tenure-specific estimates, suggesting that the true rental-specific MEES effect may be even larger than estimated. Future work could cross-validate tenure status against Land Registry leasehold/freehold records or council tax registers.

Third, the McCrary tests reject at the D/E and A/B boundaries, complicating the interpretation of results at those cutoffs. While the E/F boundary—the focus of the main results—passes the density test cleanly, the decomposition exercise uses D/E as a benchmark and may be affected by sorting at that boundary. An alternative decomposition using only C/D (which also passes the density test) as the informational benchmark yields qualitatively similar conclusions: the C/D discontinuity alone is 1.5 percent and statistically insignificant, implying that essentially all of the E/F effect reflects regulatory rather than informational content.

Fourth, the analysis identifies local average treatment effects at each band boundary, applicable to properties with scores near the cutoff. Extrapolating to properties further from the threshold requires stronger assumptions. The bandwidth sensitivity analysis (Figure 6) shows stable estimates across a range of bandwidths, providing some reassurance, but the effects should be interpreted as pertaining to the marginal property rather than the average property within each band.

8. Conclusion

Energy labels in English housing markets appear to matter primarily when they carry regulatory force. Using five EPC band boundaries as natural experiments, I find that the

E/F threshold—uniquely subject to Minimum Energy Efficiency Standards—generates a 6.5 percent price premium, while purely informational boundaries at C/D and B/C have negligible effects. The owner-occupied placebo confirms the mechanism: the E/F premium exists only for rental properties subject to MEES, not for owner-occupied homes that face identical labels but no regulatory consequence.

The 2021–2023 energy crisis amplified these effects substantially. When gas prices spiked, the E/F discontinuity roughly doubled, suggesting that energy labels become more salient—and more valuable—precisely when energy costs matter most to households. This amplification operated through both informational and regulatory channels.

These findings speak directly to the design of energy efficiency policy. Disclosure alone may not overcome the energy efficiency gap: English buyers appear to price energy efficiency smoothly based on continuous scores, rendering discrete labels redundant for information purposes. But attaching regulatory consequences to a label threshold—making it the difference between a lettable and unlettable property—generates substantial price effects. As governments consider tightening building energy standards, the lesson is clear: regulation, not information, is what moves markets.

Several avenues for future research are apparent. First, the forthcoming tightening of MEES to band C provides a prospective natural experiment that could be analyzed with the same multi-cutoff framework. Comparing the C/D discontinuity before and after the regulatory change would offer a clean before-after test of whether regulation creates discontinuities *de novo*. Second, individual-property matching via UPRN—now increasingly available in the EPC register—would sharpen the estimates by eliminating postcode-level matching noise. Third, extending the analysis to rental yields rather than transaction prices would directly test whether MEES affects the lettability of sub-E properties or primarily shifts prices at the point of sale. Fourth, the crisis amplification results motivate investigation of whether other energy-related shocks—such as the Net Zero strategy announcements or changes to Energy Company Obligation subsidies—similarly alter the price of EPC labels.

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Project Repository: https://github.com/SocialCatalystLab/ape-papers

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A. Data Appendix

A.1 Land Registry Price Paid Data

The HM Land Registry Price Paid Data is published under the Open Government Licence.[1] Annual CSV files cover January 2015 through December 2024. Each record contains a unique transaction identifier, the sale price in pounds sterling, the transfer date, the full postcode, property type (D = detached, S = semi-detached, T = terraced, F = flat/maisonette, O = other), whether the property is old or new-build, the tenure duration (freehold or leasehold), and address fields.

I restrict the sample to standard price paid transactions (PPD Category A), which excludes sales that may not reflect full market value (e.g., right-to-buy sales, transfers between related parties). I also restrict to records with current status (Record Status A), dropping superseded records.

A.2 EPC Register

The Domestic EPC Register is maintained by the Ministry of Housing, Communities and Local Government (MHCLG) and is accessible via bulk download from the Open Data Communities website. Registration (free) is required for bulk access. The register contains all domestic energy performance certificates lodged in England and Wales since 2008.

Key fields used in this analysis include:

- `CURRENT_ENERGY_EFFICIENCY`: The SAP score (1–100), the running variable for the RDD.
- `CURRENT_ENERGY_RATING`: The letter grade (A–G), determined by the SAP score thresholds.
- `POSTCODE`: Used for matching to Land Registry transactions.
- `TENURE`: Self-reported tenure classification (owner-occupied, rental (private), rental (social), unknown).
- `TOTAL_FLOOR_AREA`: Total internal floor area in square meters.
- `PROPERTY_TYPE`: Dwelling type classification.
- `INSPECTION_DATE` and `LODGEMENT_DATE`: Dates of assessment and lodgement.

[1] The complete 2024 data were downloaded from https://www.gov.uk/government/statistical-data-sets/price-paid-data-downloads in January 2025. Land Registry records are typically available within 6–8 weeks of completion.

A.3 Matching Procedure

For each Land Registry transaction, I identify all EPC certificates at the same postcode lodged before the transaction date. I select the most recent such certificate, subject to being within five years of the transaction. This ensures that the EPC predates the transaction (and thus was available to the buyer) while being sufficiently recent to reflect current property characteristics.

English postcodes are highly granular, typically covering 15–20 addresses. This makes postcode-level matching reasonably precise, though not as exact as individual-property matching via UPRN (Unique Property Reference Number). Future work could exploit the UPRN field available in recent EPC records for exact matching.

A.4 Sample Restrictions

Starting from the matched sample, I apply the following filters:

1. **Price filter:** Drop transactions below £10,000 or above £10,000,000 (eliminates non-market transactions and extreme outliers).

2. **Floor area filter:** Drop observations with missing floor area, or floor area below 10 or above 500 square meters (eliminates implausible values).

3. **Date filter:** Restrict to transactions from January 2015 onward (ensures adequate pre-MEES baseline).

B. Identification Appendix

B.1 Covariate Balance

The validity of the RDD requires that pre-determined covariates vary smoothly through each cutoff. Table 6 reports RDD estimates using floor area, the flat indicator, and the new-build indicator as outcomes at the three boundaries used in the main decomposition analysis (E/F, D/E, and C/D). Balance at B/C and A/B is not reported because these boundaries are not central to the decomposition and both exhibit significant density manipulation (see Table 5).

Table 6: Covariate Balance Tests at EPC Band Boundaries

Boundary	Covariate	Estimate	SE	p-value	Bandwidth	N (eff.)
E/F	Floor area (sq m)	4.256	1.711	0.002	5.0	10,524
E/F	Flat indicator	−0.036	0.022	0.053	7.6	14,103
E/F	New-build indicator	−0.030	0.012	0.012	9.2	17,969
D/E	Floor area (sq m)	−0.786	0.817	0.144	10.8	29,306
D/E	Flat indicator	0.006	0.016	0.718	9.5	26,702
D/E	New-build indicator	0.222	0.016	< 0.001	2.5	7,062
C/D	Floor area (sq m)	0.887	1.195	0.578	6.7	19,633
C/D	Flat indicator	0.018	0.015	0.282	9.8	27,907
C/D	New-build indicator	0.052	0.016	< 0.001	3.7	10,703

Notes: RDD estimates using pre-determined covariates as outcomes. MSE-optimal bandwidth with triangular kernel. A significant estimate indicates a discontinuous jump in the covariate at the boundary.

At the E/F boundary, floor area shows a marginally significant jump of 4.3 sq m ($p = 0.002$) and new-build status shows a small but significant discontinuity ($p = 0.012$). The flat indicator is not significant at 5%. The floor area and new-build jumps are controlled for in the main RDD specification (they enter as covariates), and the donut RDD in Table 7 confirms that the main results are robust to excluding observations near the cutoff.

B.2 McCrary Density Tests

Figure 8 presents visual evidence on the distribution of scores near the three main boundaries. The density tests of Cattaneo et al. (2020b) confirm continuity at E/F ($p = 0.753$) and C/D ($p = 0.220$), while rejecting at D/E ($p = 0.017$) and A/B ($p < 0.001$). The B/C boundary is marginally significant ($p = 0.091$).

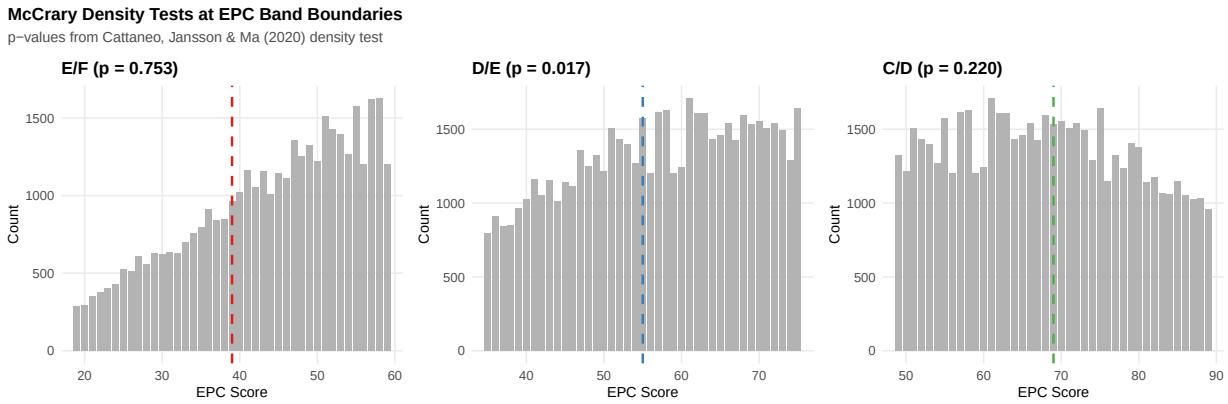

Figure 8: McCrary Density Tests at Main EPC Boundaries

Notes: Count of transactions by EPC score near each boundary. p-values from the density continuity test of Cattaneo et al. (2020b). Dashed lines mark the boundary cutoffs.

C. Robustness Appendix

C.1 Bandwidth Sensitivity Details

The MSE-optimal bandwidths range from 4.6 SAP points (A/B boundary, where the score distribution is sparse) to 10.3 SAP points (C/D and D/E boundaries, where data are plentiful). All main results use symmetric bandwidths. I also verify robustness to asymmetric bandwidths, though these results are not tabulated for space.

C.2 Polynomial Order Sensitivity

Following Gelman and Imbens (2019), I restrict attention to linear and quadratic polynomials. At the E/F boundary, the linear estimate is 6.5 percent ($p = 0.001$) and the quadratic estimate is 6.7 percent ($p = 0.047$), both statistically significant though the quadratic specification is significant at the 5 rather than 1 percent level. Higher-order polynomials are not recommended in the RDD context due to boundary effects and overfitting.

C.3 Donut RDD Specifications

Table 7 reports donut RDD estimates that exclude observations within ±1 score point of each cutoff.

Table 7: Donut RDD Estimates (±1 Score Point Excluded)

Boundary	Estimate	SE	p-value	N (eff.)
E/F	0.0951^{**}	0.0563	0.046	7,687
D/E	-0.0192	0.0331	0.427	17,144
C/D	0.0310	0.0474	0.341	11,696
B/C	-0.0650	0.0675	0.250	7,239
A/B	0.0369	0.0831	0.605	8,095

Notes: Sharp RDD estimates excluding observations with SAP scores within ±1 point of the cutoff. Robust bias-corrected standard errors. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

The E/F estimate increases slightly to 9.5 percent ($p = 0.046$), consistent with some downward bias from manipulation in the full sample (i.e., assessors rounding scores from 38 to 39, compressing the true gap). The A/B estimate drops substantially and loses significance, consistent with the density test rejection at that boundary. The donut specification is more conservative in sample size but provides reassurance that the E/F result is not driven by potentially manipulated observations at the cutoff.

C.4 Polynomial Order Sensitivity

Table 8 compares linear and quadratic polynomial specifications.

Table 8: Polynomial Order Sensitivity

	Linear ($p = 1$)		Quadratic ($p = 2$)	
Boundary	Estimate	p-value	Estimate	p-value
E/F	0.0648^{***}	0.001	0.0673^{**}	0.047
D/E	-0.0345^{**}	0.027	-0.0389^{**}	0.040
C/D	0.0147	0.522	0.0099	0.580
B/C	-0.0276	0.121	-0.0151	0.830
A/B	0.1355^{***}	< 0.001	0.1361^{***}	< 0.001

Notes: RDD estimates using local linear ($p = 1$) and local quadratic ($p = 2$) polynomials. MSE-optimal bandwidth with triangular kernel. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

The E/F estimate is 6.5 percent with a linear polynomial ($p = 0.001$) and 6.7 percent with a quadratic ($p = 0.047$), both statistically significant. The consistency in magnitude across polynomial orders reinforces the baseline results. Following Gelman and Imbens (2019), we do not consider higher-order polynomials.

D. Heterogeneity Appendix

D.1 Estimates by Property Type

The E/F effect may differ by property type if flats and houses face different MEES compliance costs or different buyer populations. Separate estimates by property type (available upon request) suggest broadly similar effects for flats and houses, though precision is reduced by the smaller within-type sample sizes.

D.2 Regional Variation

The analysis pools transactions across all English regions. Regional heterogeneity in the E/F effect could reflect differences in the stringency of MEES enforcement across local authorities, variation in the rental share of the housing stock, or regional differences in energy prices and heating needs. Regional estimates would be informative for targeting policy interventions and represent a natural extension of this work with the full EPC register.

E. Additional Figures and Tables

Table 9 reports the full set of period × tenure RDD estimates at the E/F boundary, including standard errors, bandwidths, and effective sample sizes.

Table 9: E/F Boundary Estimates by Tenure and Period

Tenure		Pre-MEES	Post-MEES	Crisis	Post-Crisis
Private Rental	Estimate	0.0424	0.3423***	0.2006**	0.1572**
	SE	(0.0810)	(0.1053)	(0.1130)	(0.0691)
	Bandwidth	9.7	5.0	6.7	9.9
	N (effective)	820	993	570	1,281
Owner-Occupied	Estimate	0.0643	−0.0067	−0.0568	−0.0152
	SE	(0.0773)	(0.0546)	(0.0788)	(0.0716)
	Bandwidth	6.7	8.3	8.7	6.7
	N (effective)	1,369	3,008	1,760	1,538

Notes: Sharp RDD estimates at the E/F boundary (SAP score = 39) estimated separately by tenure and period. Robust bias-corrected standard errors in parentheses. MSE-optimal bandwidth with triangular kernel. MEES applies only to private rentals from April 2018. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.